J.17

NAME ☞	
SCHOOL ☞	
TEACHER ☞	

Pre-Junior Certificate Examination, 2015

Mathematics

Paper 1

Ordinary Level

Time: 2 hours

300 marks

For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5		15	
6			
7			
8			
9			
10		Total	

School stamp

Running total	

Grade

Instructions

There are 15 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times, you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

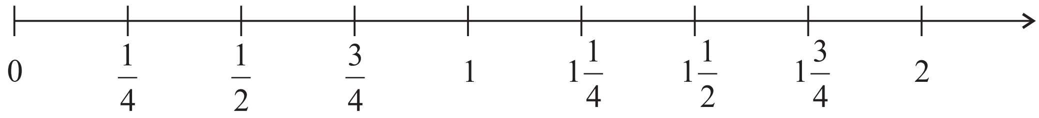
You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

Part of a number line is shown in the diagram below.



(a) **(i)** Which number on the number line above has the same value as $0{\cdot}75$?

(ii) Which number on the number line above has the same value as $\frac{3}{2}$?

(iii) Which number on the number line above is half-way between 0 and $1\frac{1}{2}$?

(b) Show the approximate position of $\frac{1}{3}$ on the number line above.

(c) Write as a fraction the number that is half-way between $\frac{1}{3}$ and 1.

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Question 2 **(Suggested maximum time: 5 minutes)**

The table shows the number of tickets sold for a pop concert on five consecutive nights.

Night	Tickets sold
1	41 378
2	20 672
3	21 338
4	41 685
5	33 587

(a) Find the difference between the night with the highest number of tickets sold and the night with the lowest number of tickets sold.

(b) For which night was the number of tickets sold closest to 21 000?

(c) 2441 of the tickets sold for the concert on the last night were **not** used.

How many people attended the concert that night? Write your answer correct to the nearest hundred.

(d) It was reported in a newspaper that 160 000 tickets were sold in total between all of the concerts. Explain how the newspaper came to this conclusion.

Question 3 **(Suggested maximum time: 5 minutes)**

(a) **(i)** Write $0{\cdot}3$ as a percentage.

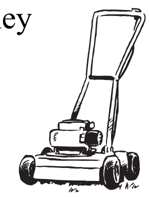
(ii) Find $\frac{3}{7}$ of 56.

(iii) Write 42% as a fraction.
Give your answer in its simplest form.

(b) Insert brackets to make the calculation below correct.

$$18 - 6 \div 2 + 5 = 11$$

(c) Sean and Brian did some gardening work together. They decided to share the money they were paid in the ratio of the number of hours each of them worked.
Sean worked 5 hours and Brian worked 7 hours.
The boys were paid a total of €76·20.



How much did each boy receive?

Sean =

Brian =

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Question 4 **(Suggested maximum time: 10 minutes)**

Liam has €6000 and would like to invest it for two years. He is considering the special savings accounts in two banks, which are shown below.

Bank A

Invest €6000

for 2 years

Compound Interest at 2% per annum

Any withdrawals within the investment period will reduce the amount of interest earned.

Bank B

Invest €6000

for 2 years

Compound Interest

Total interest guaranteed €303·75

No withdrawals allowed within the investment period.

(a) Find the total amount of Liam's investment at the end of the first year if he chooses the special savings account in Bank A.

(b) Find the total amount of interest Liam would earn over the two years if he chooses Bank A.

(c) Show that Bank B is offering an interest rate of 2·5% per annum on its special savings account.

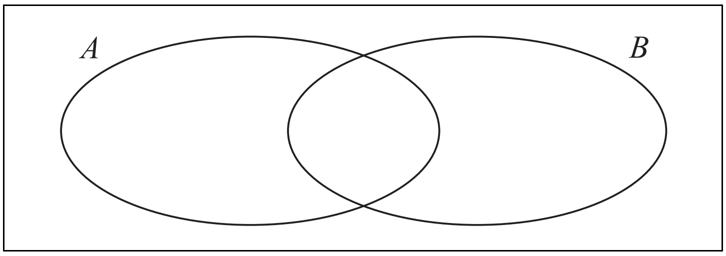
(d) In which bank should Liam invest his money? Give a reason for your answer.

Answer:

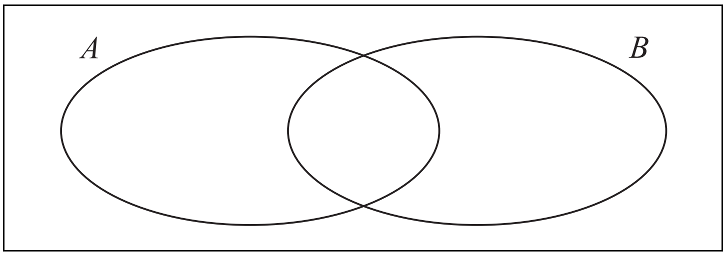
Reason:

Question 5 **(Suggested maximum time: 5 minutes)**

(a) **(i)** In the Venn diagram below, shade the region that represents $A \cup B$.



(ii) In the Venn diagram below, shade the region that represents $B \setminus A$.



(b) $X = \{0, 1\}$.

(i) Write down $\#X$.

(ii) Jack claims that X has 4 subsets. Is he correct?
Explain your answer by listing all the elements of each of the subsets of X.

Answer:

Explanation:

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Question 6 **(Suggested maximum time: 5 minutes)**

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
F is the set of factors of 10. P is the set of prime numbers, from 1 to 10 inclusive.

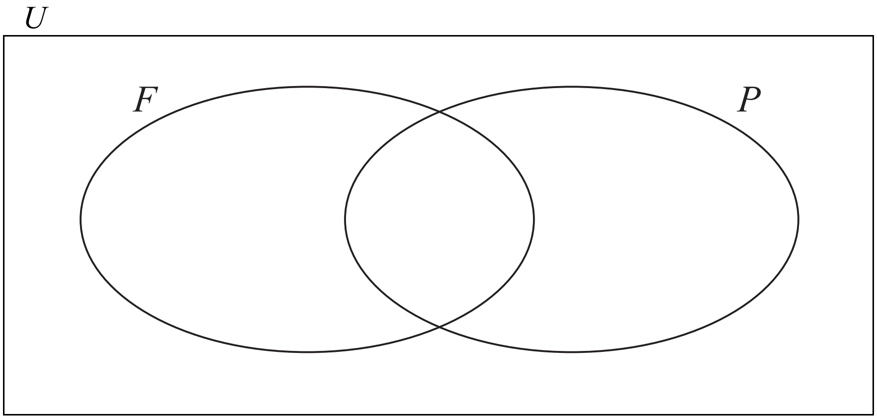
(a) **(i)** List the elements of the set F.

$$F = \{ \quad , \quad , \quad , \quad \}$$

(ii) List the elements of the set P.

$$P = \{ \quad , \quad , \quad , \quad \}$$

(b) Fill in the Venn diagram below by placing all elements of U in the correct regions.



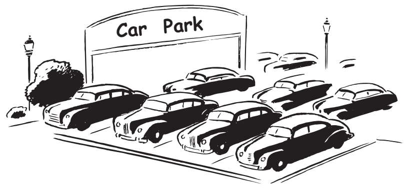
(c) **(i)** List the elements of $(F \cup P)'$, where $(F \cup P)'$ is the complement of the set $F \cup P$.

$$(F \cup P)' = \{ \qquad\qquad \}$$

(ii) Describe in your own words what $(F \cup P)'$ represents.

Question 7 **(Suggested maximum time: 5 minutes)**

The parking charges in a car park are shown below.



City Car Park

Pay & Display

€1·60 per hour

Pay Machines only take coins

(a) Ben paid €4·80 to use the car park.
For how long did he pay?

(b) The pay machines in the car park take the following coins:

Ben used exactly 6 coins. He did not use any 5c coins.
Show **three** possible ways in which he could have paid.

Answer 1: ______, ______, ______, ______, ______, ______.

Answer 2: ______, ______, ______, ______, ______, ______.

Answer 3: ______, ______, ______, ______, ______, ______.

(c) The car park offers users a discount of 20c per hour at weekends.
Find the percentage discount offered at weekends.

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Question 8 **(Suggested maximum time: 5 minutes)**

(a) **(i)** Write down the value of 2^4.

(ii) Find the value of the square root of 36.

(iii) Write $10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$ in the form of 10^x, where $x \in \mathbb{N}$.

(b) The table below shows expressions of x where x is raised to certain powers.

$\frac{x^9}{x^2}$	$(x^3)^3$	x^4	$2x^3$
$x^3 + x^3$	x^{12}	$x^3 \times x^7$	x^7
x^9	$x^{10} \div x^6$	x^{10}	$(x^6)^2$

Match up pairs of expressions that are the same from the above table.
One pair is completed for you.

(i) x^{12} and $(x^6)^2$

(ii) and

(iii) and

(iv) and

(v) and

(vi) and

Question 9 (Suggested maximum time: 10 minutes)

(a) **(i)** Select the expression from the table below that is equivalent to $\mathbf{3 \times \boldsymbol{m}}$.

$m \times m \times m$	$3m$	m^3	3^m

Expression: ______________

(ii) Select the expression from the table below that is equivalent to $\boldsymbol{n \times n \times n \times n}$.

n^2	$4n$	$2n^2$	n^4

Expression: ______________

(iii) Select the expression from the table below that is equivalent to $\boldsymbol{x + y}$.

xy	$y + x$	yx	$2xy$

Expression: ______________

(b) **(i)** Simplify $3p + 2q - p + 5q$.

(ii) Expand and simplify $5(a + 4) + 2(6a - 1)$.

(c) Daniel knows the value of $\boldsymbol{x}$ is **6** or **7** and the value of $\boldsymbol{y}$ is **3** or **4**.
Evaluate the largest and smallest possible values of $\mathbf{3\boldsymbol{x} - 2\boldsymbol{y}}$.

Smallest value = ______________ Largest value = ______________

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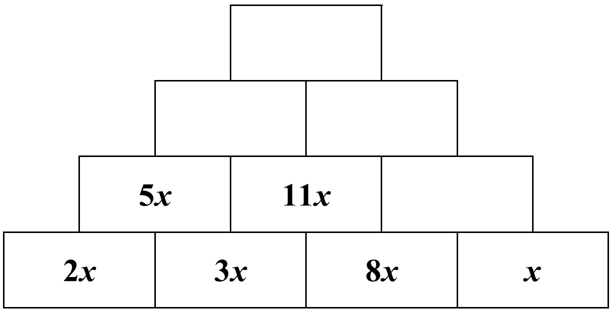
Question 10 **(Suggested maximum time: 10 minutes)**

(a) In an algebra pyramid you add the two expressions in the lower blocks to find the expression in the block above (for example, $2x + 3x = 5x$).

Complete the algebra pyramid below by filling in the empty spaces.



(b) Factorise fully each of the following:

(i) $12xy - 3y$

(ii) $x^2 + 2x - 48$

(iii) $x^2 - 81$

(c) Using your answer from part **(b)(iii)**, or otherwise, solve the equation $x^2 - 81 = 0$.

Question 11 **(Suggested maximum time: 10 minutes)**

(a) **(i)** Find the next three terms of the sequence below.

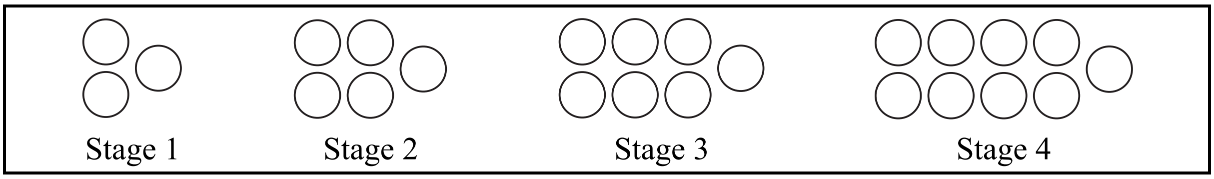
0, 3, 6, ______, ______, ______

(ii) Suggest a rule for continuing the sequence below.

41, 37, 33, 29, ……

(iii) The sequences in parts **(i)** and **(ii)** continue.
Find the first term that is common to both sequences.

(b) The first four stages of a pattern are shown below.

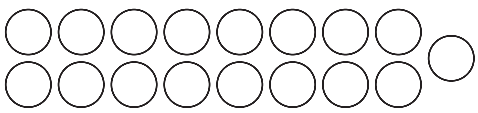


(i) Tick the correct box to complete the sentence.

The number of discs in each pattern in the sequence is…

always even ☐ always odd ☐ either even or odd ☐

(ii) Shade in the appropriate discs below to show Stage 6 of the pattern.



(iii) Fill in the two boxes below to complete the rule to determine the number of discs in each stage of the pattern.

Number of discs = Stage number × ☐ + ☐

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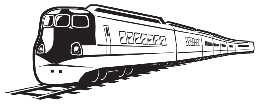
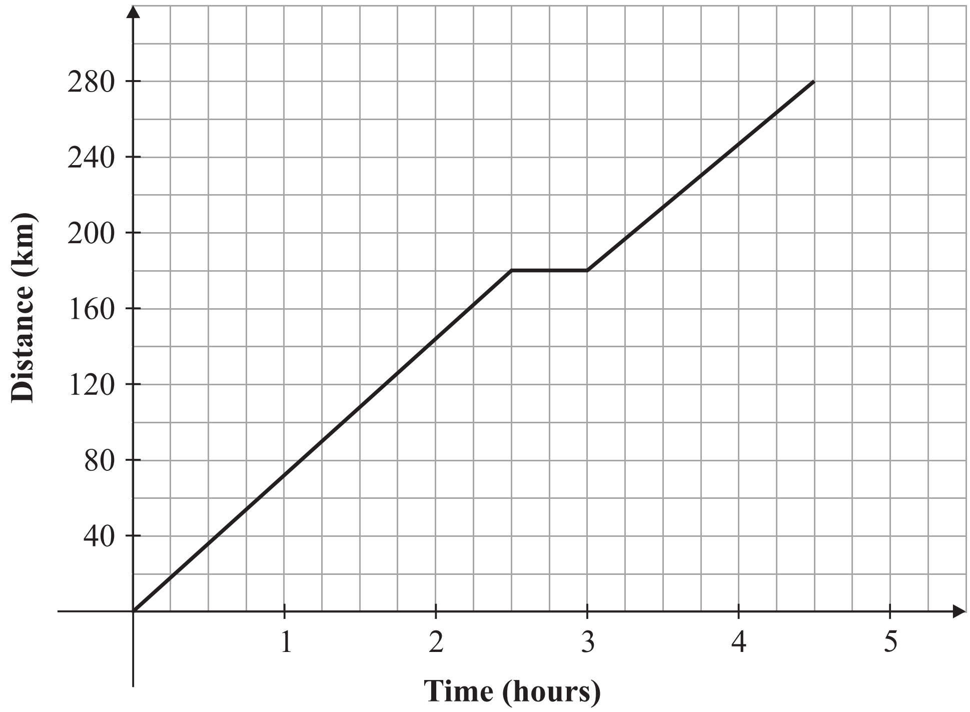
Question 12 **(Suggested maximum time: 5 minutes)**

Sandra travelled by train from Dublin to Tralee. The train stopped in Mallow and waited for passengers from a connecting train. The graph below shows the distance and journey time from Dublin.





(a) From the graph, what is the distance from Dublin to Tralee by train?

(b) Find the length of time the train was moving.

(c) Sandra said that the train travelled faster from Dublin to Mallow than from Mallow to Tralee. Is she correct? Give a reason for your answer.

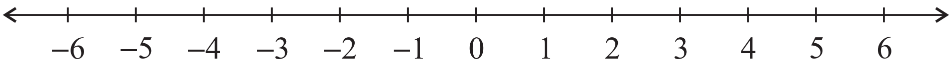
Answer:

Reason:

Question 13 **(Suggested maximum time: 5 minutes)**

(a) **(i)** Solve the inequality $5x - 4 \leq 16$, where $x \in \mathbb{N}$.

(ii) Graph your solution to part **(i)** on the number line below.



(b) Solve the simultaneous equations

$$2x + 3y = 11$$
$$4x - y = 1.$$

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Question 14 **(Suggested maximum time: 10 minutes)**

$\{(2, 3), (4, 6), (6, 9), (8, 12)\}$ are four couples of a function f.

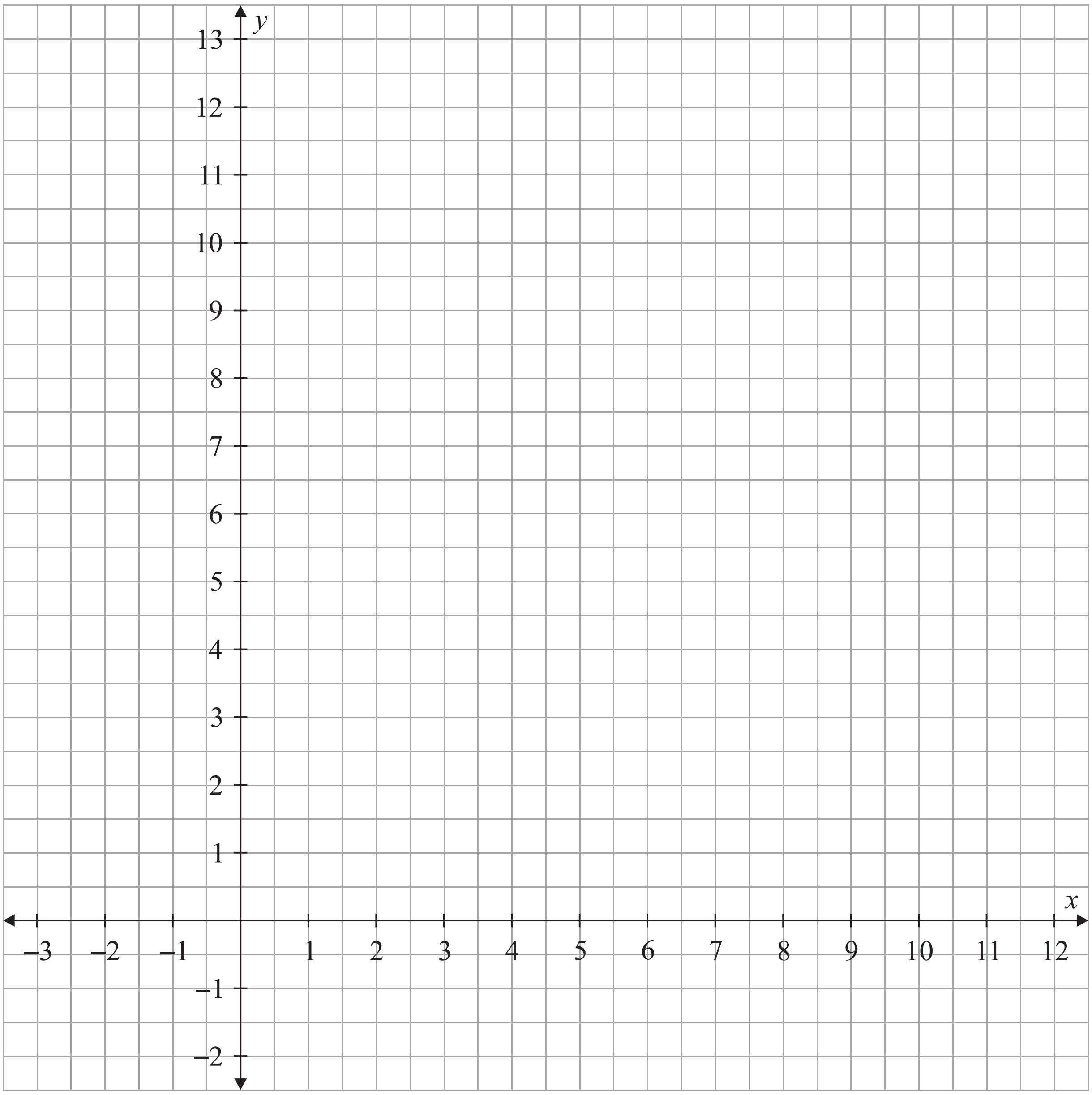
(a) **(i)** Write down the domain and the range of the function f.

Domain $= \{ \quad , \quad , \quad , \quad \}$

Range $= \{ \quad , \quad , \quad , \quad \}$

(ii) Explain the relationship between the domain and the range of the function f.

(b) Plot the four couples of f on the diagram below.



(c) John says that (2·5, 4) is also a couple of the function f.
Using the diagram in part **(b)**, or otherwise, show why he is incorrect.

(d) Using your diagram in part **(b)**, write down two other couples of f.

(e) Barry claims that {(2, 5), (3, 6), (4, 7), (4, 8)} are couples of a function g.
Is he correct? Give a reason for your answer.

Answer:

Reason:

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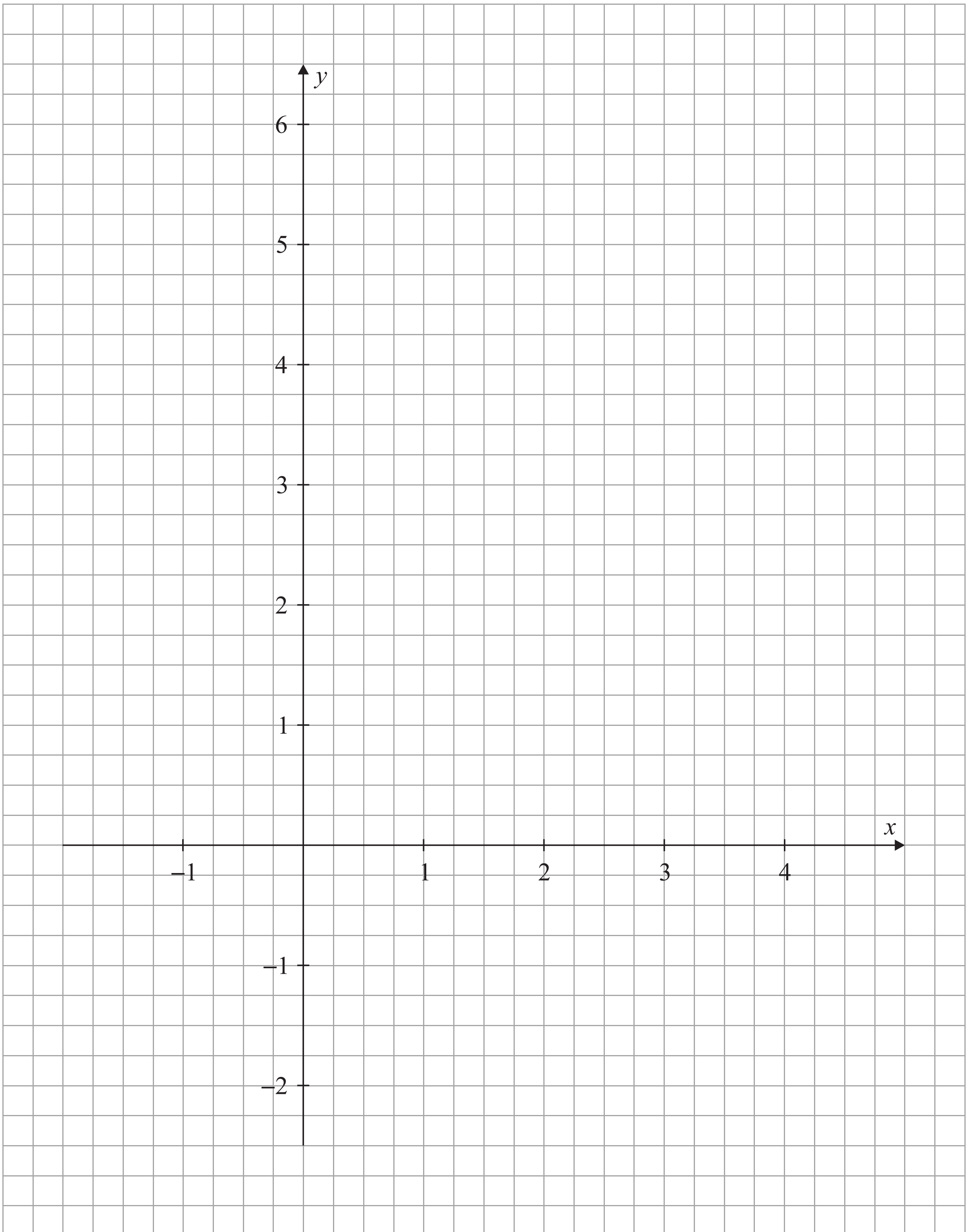
Question 15 **(Suggested maximum time: 15 minutes)**

(a) Draw the graph of the function $f : x \mapsto x^2 - 3x + 1$ in the domain $-1 \leq x \leq 4$, where $x \in \mathbb{R}$.

There is more room for working out on the next page.



The function $f : x \mapsto x^2 - 3x + 1$ models the predicted air temperature, in **degree Celsius** (**°C**), at Cork Airport over a 10-hour period of time.

The x-axis represents the time from 10 p.m. ($x = -1$) to 8 a.m. ($x = 4$).

Use your graph from part **(a)** to answer the following questions. Show your work on the graph.

(b) Find the coldest predicted air temperature at Cork Airport.

(c) Between what times is the air temperature at the airport predicted to be below freezing (0°C)?

(d) Frequently during wintertime, flights to and from Cork are suspended when the air temperature falls below 2·5°C as fog forms, reducing visibility at the airport.
According to your graph, for how long will flights to and from Cork be suspended?

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Pre-Junior Certificate, 2015 – Ordinary Level

Mathematics – Paper 1

Time: 2 hours

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